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Muscle coactivation levels reflect a tradeoff between metabolic cost and performance when responding to physical disturbances during upper limb posture control.

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SUMMER RESEARCH LITERATURE REVIEW

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Abstract

Muscle coactivation refers to the simultaneous activity of muscles that have opposing functions. Recent observations suggest that coactivation may improve performance by making the nervous system more responsive to sensory feedback. Muscle activity is costly, however, and we know little about the metabolic cost of coactivating muscles. This study investigates the trade-off between metabolic cost and performance under different levels of muscle coactivation. Five healthy participants attempted to maintain a fixed arm posture in a Kinarm exoskeleton robot while encountering mechanical disturbances. The disturbances rapidly extended or flexed the elbow and displaced the participant's arm from the target. They were required to return to the target within 400 ms. Participants first performed the task under self-selected levels of muscle coactivation. Biofeedback was then used to maintain a level of coactivation corresponding to [0.5, 2, 4, 8, 16]x their self-selected (1x) level. Metabolic cost was measured with gas respirometry, the activity of upper limb muscles using surface EMG, and behavioral responses using success rates. At self-selected levels of coactivation, participants had a success rate of 30-50%. We found that muscles became more responsive to the disturbances at higher coactivation levels, which limited the displacement of the arm, improved success rates to nearly 100% but increased metabolic costs. The findings suggest that participants prioritize metabolic cost over performance at self-selected levels of muscle coactivation. This research offers insights into the role of muscle coactivation in responding to sensory feedback.

Literature Review

Introduction

Imagine a sprinter at the starting line of a competition; their muscles are tense and ready for action. This is an example of muscle coactivation because the sprinter's hamstrings and quadriceps, two opposing muscles in their legs, are cooperating to allow their body to control human movement. Coactivation helps to stabilize our joints, control how stiff our limbs are, and fine-tune our movements, which are important aspects in sports but also in everyday life (Kesar et al., 2019; Vieira et al., 2019). People have historically thought that the main job of coactivation was to make muscles stiff and resistant to movement, and this idea is well supported by research (Vieira et al., 2019). However, recent studies have shown that there is more to coactivation especially in situations like sprinting where activating your opposing muscles seems to have a broader role in how we move (Kesar et al., 2019).

The connection between how well people perform and muscle coactivation is complex and not completely understood. More muscle coactivation generally means better joint stability, but it also uses more energy, which can impact performance (Molina-Rueda et al., 2023). In sports and rehabilitation, coactivation can help people make fewer mistakes, improve coordination, and respond better to their feelings (Antohe et al., 2020). This literature review will look into what we know about coactivation, particularly how it works and why it matters in different situations such as sports or everyday activities. By looking at the balance between muscle coactivation, metabolic cost and performance we hope to shed some light on its complex role in human movement.

Physiological Mechanisms of Muscle Coactivation

Muscle coactivation is influenced by the control strategies of the nervous system and can vary in different levels of intensity or activation (Nagai et al., 2011). The control strategies include both the feedback and feedforward processes (Law & Avin, 2013). The feedforward mechanism prepares the body by activating the muscle for expected movements or perturbation, while the feedback mechanism involves fixing the muscle activity in response to sensory inputs such as an external force that causes a change in the position of the joint (Pienciak-Siewert et al., 2020). These mechanisms are very critical for maintaining human balance and doing precise movement, specifically in an unpredictable environment (Nagai et al., 2011).

Research has shown that the central nervous system (CNS) can control the level of coactivation based on our task demands, such as the need for stability or responding rapidly to mechanical perturbation (CF, 2017). For example, during a high-precision task our CNS increases our level of coactivation to minimize our movement to enhance our performance (Balshaw et al., 2018). To increase the level of muscle coactivation, the CNS utilizes a complex motor planning system that integrates sensory information and predicts the outcomes of the actions, which allows for an effective response to the environmental challenge (Law & Avin, 2013). Furthermore, research has shown that patterns of muscle coactivation can differ between populations. For example, age-related variations have been observed, with older adults frequently showing higher levels of muscle coactivation as a means of delaying the deterioration of their muscle strength and coordination (Nagai et al., 2011).

Muscle coactivation is also known to be organized into a group of muscles that can be activated together to produce a coordinated movement, which is known as muscle synergies (Kutch & Valero-Cuevas, 2012). Muscle synergy can help simplify the control of complex movements, and the nervous system will be able to coordinate actions more efficiently, by balancing the trade-off between performance and metabolic cost (Vieira et al., 2019). For example, studies related to electromyography (EMG) have shown different patterns of muscle activation which relate to various movements, such as reaching, grasping, and walking (Boudarham et al., 2016). These muscle synergies not only improve our motor performance but also improve the coordination of the muscle, which influences the overall energy expenditure (Le, 2023). Thus, understanding muscle synergies is very important to analyze how the CNS can control different levels of coactivation to achieve task-based goals, which can help in motor control and rehabilitation strategies (Le, 2023). For example, in a rehabilitation setting, understanding how muscle coactivation works can help create a medical intervention that can improve motor function in patients with an impairment related to the nervous system (Dyer et al., 2011).

Modulating Joint Stiffness Through Coactivation

The primary function of muscle coactivation is to control joint stiffness (Vieira et al., 2019). The body increases the stiffness of the joint through the coactivation of the agonist and antagonist muscles, which makes the joint more resistant to external perturbation (Gebel et al., 2019). Increasing the stiffness of the joint helps maintain joint stability, which is very helpful during tasks that apply a sudden force (Gebel et al., 2019). A person standing in a balanced position is an example of how coactivation of the ankle and knee muscles can help stabilize the body from swaying (Lévénez et al., 2005). A recent study has also shown that large levels of coactivation, which leads to a greater joint stiffness, reduce the risk of injuries during a dynamic activity (Molina-Rueda et al., 2023). This protective mechanism is primarily seen when the body has to respond to an unexpected lateral perturbation, increasing the overall stability of the joint (Vieira et al., 2019).

The role of muscle coactivation is seen across a wide range of motor tasks which includes maintaining the stability of the posture to executing rapid movements (Pienciak-Siewert et al., 2020). Muscle coactivation can be categorized into three main branches according to its roles which are stability enhancement, response optimization and movement precision (CF, 2017). The most popular role of muscle coactivation is associated with enhancing stability and balance (Deffeyes et al., 2012). For activities such as standing, walking, or carrying objects, maintaining balance requires a constant adjustment to the body's center of mass (COM). These adjustments to the body's COM are carried through muscle coactivation that provides a stable platform from which movements can be initiated (Deffeyes et al., 2012). For example, a person standing still is achieved by the coactivation of the calf and shin muscles to maintain ankle stability, reducing the risk of falling and swaying (Ting & Macpherson, 2005).

Many fine motor tasks, which include writing or working with small objects require a high level of precision that can be achieved by higher levels of muscle coactivation (Law & Avin, 2013). A higher level of coactivation results in a reduction in the variability in movement and allows for more controlled movements because of increased joint stiffness (Balshaw et al., 2018). This reduction in the variability in movement is very important in real-life scenarios which require a precise hand movement such as surgery or playing musical instruments (Kutch & Valero-Cuevas, 2012). Furthermore, muscle coactivation plays a role in preparing the human body for fast or rapid responses to sensory feedback (Song et al., 2003). An example of this response is shown in activities like sprinting where athletes coactivate their muscles before the starting signal, this prepares the muscles and nervous system for a quick and forceful start (Song et al., 2003).

Variability in Coactivation Across Individuals and Contexts

Muscle coactivation depends on multiple factors such as age, skill level, and the demands of the task (Nagai et al., 2011). Since coactivation differs across an individual and its contexts, understanding this variation can provide insight into how coactivation is used to improve performance and control energy expenditure (Peterson & Martin, 2010). For example older adults show higher levels of muscle coactivation compared to younger adults for tasks that require balance and stability (Nagai et al., 2011). This increased coactivation serves as a mechanism to reduce the age related decline in body coordination and muscle strength (Nagai et al., 2011). Although this increased muscle coactivation also results in increased metabolic cost, which can cause fatigue and reduced endurance in older adults (Peterson & Martin, 2010). Age can therefore significantly affect the extent and pattern of muscle coactivation.

A person skilled in a particular activity can influence muscle coactivation patterns. Professional athletes display coactivation strategies that reduce unnecessary muscle activity while maximizing performance (Balshaw et al., 2018). An experienced pianist for example uses minimal coactivation during performance to allow for greater fluidity in their movement (Pereira & Gonçalves, 2011). On the other hand, a beginner playing the piano shows a higher level of coactivation as they try to stabilize their movements and reduce errors (Pääsuke et al., 2000). The task demands also play an important role in controlling the level of muscle coactivation. For tasks that need a lot of stability, like lifting a heavy object or keeping a steady posture, higher levels of muscle coactivation are used (Peterson & Martin, 2010). In contrast, tasks that are dependent on speed and fluidity may need lower coactivation levels to allow for more range of motion (ROM) and flexibility (Peterson & Martin, 2010). This variation in coactivation based on the task shows how the nervous system adapts to maximize performance.

Balancing Performance and Metabolic Cost

Muscle coactivation plays a pivotal role in affecting metabolic cost. While it can boost our performance by improving stability and responsiveness as explained above, it also raises the body's energy demand (Peterson & Martin, 2010). Coactivation increases overall muscle activity, which leads to higher energy use (Peterson & Martin, 2010). Increased metabolic cost can be measured with methods like gas respirometry, which checks how much oxygen is being consumed and how much carbon dioxide is produced (Peterson & Martin, 2010). Studies show that tasks requiring a lot of muscle coactivation, like holding a steady posture against a force, use more energy than tasks with less coactivation (Peterson & Martin, 2010).

People often choose the level of muscle coactivation by looking at the benefits of increased stability and performance against the energy cost (Peterson & Martin, 2010). Looking at a task that requires a rapid response to visual stimuli, people generally increase their coactivation to reduce their reaction time at the expense of a higher energy cost (Peterson & Martin, 2010). However, for a task that has a lower performance demand, an individual may choose lower levels of muscle coactivation to conserve energy (Peterson & Martin, 2010). Thus, understanding the trade-off between muscle coactivation, metabolic expense and performance is important for maximizing performance in both athletic and rehabilitation contexts.

Interplay Between Muscle Coactivation and Sensory Feedback

Muscle coactivation plays a very important role in the control of neural sensory feedback. Recent studies suggest that increasing muscle coactivation enhances how the CNS responds to external disturbances. For instance, Kesar et al. (2019) discovered that the excitability of corticomotor pathways for ankle muscles increases when there is coactivation between agonist and antagonist muscles. Increased excitability or responsiveness helps maintain balance and stability during dynamic activities (Boudarham et al., 2016). Thus, coactivation not only improves joint stability but also helps prepare the CNS for fast adjustments in response to sensory inputs. This becomes very beneficial during tasks that require quick reactions to physical disturbances. The ability of the CNS to control coactivation levels based on task demands, suggests the importance of understanding this relationship in the context of motor control.

Moreover, the concept of muscle coactivation as a compensatory mechanism is particularly important for people with motor control problems. Looking at people with multiple sclerosis, which is a progressive neurodegenerative disease of the CNS that is caused by axonal degeneration and demyelination, Molina-Rueda et al (2023) found that they show high levels of coactivation to maintain the stability during gait. This suggests that coactivation of the lower limbs during gait for patients with multiple sclerosis can result in increased joint stiffness and help to counteract the effect of external perturbations (Boudarham et al., 2016). As increased coactivation improves joint stability, it also increases metabolic costs, which emphasizes a trade-off between performance and energy expenditure (Peterson & Martin, 2010). Understanding coactivation patterns in various populations can help us learn how the brain processes sensory information and controls movement.

Different research studies on the primary motor cortex further explain the neural mechanisms behind coactivation and sensory feedback, showing how the brain organizes muscle use for different tasks. For example, Balshaw et al. (2018) explain that different patterns of muscle coactivation correspond to specific motor tasks. This indicates that the CNS is always organizing muscle use and adjusting coordination based on what each task demands. This organization really boosts movement efficiency because the CNS can tweak coactivation levels to enhance performance while also keeping energy use low (Gebel et al., 2019). Therefore, understanding how muscle coactivation interacts with sensory feedback helps us see how people adapt their motor strategies to different demands, especially in rehabilitation settings where the goal is to help individuals regain their motor skills (Kutch & Valero-Cuevas, 2012).

Implications for Rehabilitation and Clinical Applications

Muscle coactivation has important uses in rehabilitation and medical settings. Some conditions, like muscle spasticity, hyperreflexia, and motor problems, show unusual muscle coactivation patterns (Dyer et al., 2011). Understanding these patterns helps doctors create treatments to improve motor skills and reduce abnormal muscle activity (Dyer et al., 2011). For example, muscle spasticity means increased muscle tone and reflexes (Dyer et al., 2011). This happens because of too much muscle coactivation, which makes movement hard and causes joint stiffness (Dyer et al., 2011). Treatments like botulinum toxin injections, physical therapy, and neuromodulation are used to lower the excessive coactivation and help improve motor function (Dyer et al., 2011).

Patients with motor impairments which includes stroke survivors show patterns of coactivation that may be disrupted (Dyer et al., 2011). This can lead to difficulties in movement coordination (Dyer et al., 2011). So rehabilitation programs that specifically focus on fixing the coactivation patterns can help fix movement patterns in these patients (Dyer et al., 2011). Some techniques such as biofeedback control, strength training and task-specific exercise can be used to change the coactivation level and improve functional outcomes (Dyer et al., 2011). Finally, understanding the neural mechanism related to coactivation can improve the development of more effective rehabilitation strategies.

Future Research Directions

Studying muscle coactivation is an area that's always changing, with lots of chances for future research. There are new ways to look at how the brain and body work with coactivation, like using advanced imaging and neuromodulation techniques (Pereira & Gonçalves, 2011). In order to understand the range of scenarios in which coactivation can be useful, research should focus on a larger of populations, which includes older adults, athletes, and persons with impairments. Also, looking into how coactivation relates to things like muscle fatigue can help us understand motor control better.

Conclusion

Muscle coactivation is important for controlling our movement which helps with stability, accuracy, and responsiveness across many tasks. The brain controls this by complex interactions between feedforward and feedback processes, adjusting coactivation levels based on what the task needs and the person’s abilities. In conclusion, understanding muscle coactivation is important for maximizing our performance, managing energy expenditure, and creating better rehab strategies for people with movement issues.

Methodology

Kinarm Exoskeleton Adjustment:

Healthy participants were asked to place their arms in the troughs of the Kinarm exoskeleton robot which limited movement in a single plane, allowing only horizontal motions. We adjusted the device to ensure that participants could move smoothly without restrictions. We also adjusted the chair height so their arms were level with the exoskeleton arms, with feet flat on the footrest and back supported. The shoulder indicator light was turned on to ensure their shoulders were level, and aligned with the acromion process. The armrests were fine-tuned to support their arms comfortably, with elbows slightly bent, and aligned the exoskeleton arms with the participant's arm length to ensure the joints matched the shoulder, elbow, and wrist. In addition, the Kinarm device was connected to a TV screen that displayed the targets. We closed a shield so participants could not see their hands, only the visual representation of their hands as a white cursor on the screen.

Participant Preparation

Muscle activity was measured using EMG electrodes placed on specific muscles. If necessary, we shaved small areas using disposable razors to improve the signal quality. We cleaned the muscles with an alcohol swab and a reference electrode was placed on the lateral epicondyle while the EMG electrodes were placed on the brachioradialis, triceps lateralis, biceps (short) and triceps (longus) muscles. If it was their first time, we created a new subject profile in the EMG computer and recorded their age, height, and weight before fitting the VO2 mask. Participants also wore a VO2 mask which covered their nose and mouth and this measured their metabolic activity while performing the task. They were asked to minimize speaking during the task to avoid affecting their breathing rate. The machine was calibrated, and participants were informed that they would only see the white cursor aligned with their index finger after calibration.

First Task Protocol:

Participants were told that a small target would appear at the center of the screen, and they needed to move the white cursor representing their hand to the center of the target. They were asked to maintain a fixed arm posture while the robot introduced a mechanical disturbance that rapidly extended or flexed the participants' elbows. This displaced their arm from the center of the target position and participants were required to return their arm to the target within 400 milliseconds. Feedback was provided based on their performance: a green target indicated success, while a blue target indicated they were too slow or inaccurate. After each task, participants were given a brief break. During the experiment, we recorded the time and reference EMG values, using the first condition to determine the reference EMG values for the biceps with MATLAB.

Second Task Protocol:

For the second task, participants were instructed to bring the white cursor into the center of a circular target. A slider (magenta in color) and biofeedback target (green rectangular box beside the circular target) were also displayed. They were told to match the slider, representing muscle activation, to the biofeedback target, which showed the required muscle activation for that trial. The machine would then apply a force, and participants had to return the cursor to the center of the circular target. If the hand get back into the centre of the circular target within the time limit, then the target would appear green and if it was done within time, then the target would appear blue. This task was performed until failure.

Final Task:

In the final task, lasting about 2 minutes, participants were instructed to bring the white cursor into the center of the circular target and hold that position while relaxing as much as possible, even as a small load was applied. After the task, we saved the data, and participants were compensated with $20.

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